

$$\hat{P} |X\rangle \otimes |Y\rangle = |Y\rangle \otimes |X\rangle$$

$$\left[\begin{array}{l} |1\rangle \in \mathcal{H}_F \\ |2\rangle \in \mathcal{H}_B \end{array} \right. \left. \begin{array}{l} \frac{|a\rangle \otimes |b\rangle - |b\rangle \otimes |a\rangle}{\sqrt{2}} \\ \frac{|a\rangle \otimes |b\rangle + |b\rangle \otimes |a\rangle}{\sqrt{2}} \end{array} \right\} |a\rangle \otimes |b\rangle$$

$$\Phi_{\pm}, \bar{\Phi}_{\pm}, \Xi_{\pm}, \gamma_{\pm}, \dots$$

Diagram showing arrows between fields: $\Phi_{\pm} \leftrightarrow \bar{\Phi}_{\pm} \leftrightarrow \Xi_{\pm} \leftrightarrow \gamma_{\pm}$ with green circular arrows on each field and orange curved arrows between them.

$$\underline{\Phi_+} \xleftrightarrow{\text{Relabel}} \underline{\Phi_-}$$

$$\hat{O} = \hat{A} \otimes 1 \otimes 1 + 1 \otimes \hat{A} \otimes 1 + 1 \otimes 1 \otimes \hat{A} \quad \Leftarrow$$

$A \otimes B \otimes C + \dots$

$$\hookrightarrow \langle \Phi_- | \hat{O} | \Phi_- \rangle = \langle \Phi_+ | \hat{O} | \Phi_+ \rangle$$

$$\hat{P} = |\Phi_+ \times \Phi_+| + |\Phi_- \times \Phi_-| - |\bar{\Phi}_+ \times \bar{\Phi}_+| - |\bar{\Phi}_- \times \bar{\Phi}_-|$$

$$\langle P \rangle = +1 \rightarrow \Phi_{\pm}$$

$$\langle P \rangle = -1 \rightarrow \bar{\Phi}_{\pm}$$

$$\langle P \rangle = 0 \rightarrow \text{Bosonic/Fermionic}$$

$$|\Phi_+ \times \Phi_+| - |\bar{\Phi}_- \times \bar{\Phi}_-| \Rightarrow \times \text{ identical Particles!}$$

$$x, y, z, \quad p_x, p_y, p_z$$

$$\begin{aligned} \hat{y} &\rightarrow 1 \otimes \hat{y} \otimes 1 \\ \hat{x} &\rightarrow \hat{x} \otimes 1 \otimes 1 \end{aligned}$$

$$\Rightarrow A = xy(z p_z + p_z z) + zx(y p_y + p_y y) + yz(x p_x + p_x x)$$

SHO : $X_{mn} = \left(\sqrt{n+1} \delta_{m,n+1} + \sqrt{n} \delta_{n,m+1} \right) / \sqrt{2} = \langle m | \hat{X} | n \rangle$

$$P_{mn} = \left(\sqrt{n+1} \delta_{m,n+1} - \sqrt{n} \delta_{n,m+1} \right) i / \sqrt{2}$$

$$\frac{a+a^\dagger}{2} = X$$

$$\frac{a-a^\dagger}{2i} = P$$

$$(X P_x + P_x X) = i \left(\sqrt{m(n+1)} \delta_{m,n+2} - \sqrt{n(m+1)} \delta_{n,m+2} \right)$$

$|\psi\rangle : |m\rangle, |n\rangle, |s\rangle \longrightarrow |0\rangle, |1\rangle, |2\rangle \longrightarrow |\Phi_+\rangle$

$$\langle A \rangle \quad |\Phi_+\rangle := \frac{|mns\rangle + \omega |ns m\rangle + \omega^* |smn\rangle}{\sqrt{3}} \quad \omega^* = \omega^{-1} = \omega^2$$

$$\langle A \rangle = \frac{\langle mns| + \omega^* \langle ns m| + \omega \langle smn|}{\sqrt{3}} A \frac{|mns\rangle + \omega |ns m\rangle + \omega^* |smn\rangle}{\sqrt{3}}$$

$$= \frac{1}{3} \left(A_{mns, mns} + A_{ns m, ns m} + A_{smn, smn} + \omega A_{mns, ns m} + \dots \right)$$

$A_{mns, mns} \leftarrow A_{smn, mns}$

$$\langle A \rangle = A_{mns, mns} + \omega A_{mns, ns m} + \omega^* A_{mns, smn} \Rightarrow \text{Cluster inseparability}$$

$$X_{01} = 1/\sqrt{2}, \quad X_{12} = 1$$

$$(X P_x + P_x X)_{02} = -i\sqrt{2}$$

$$\langle \Phi_\pm | A | \Phi_\pm \rangle = -3\sqrt{3}$$

$$\langle \Phi_\pm | A | \Phi_\pm \rangle = 3\sqrt{3}$$

$$\langle A \rangle = 0$$

$$\rho = \frac{1}{2} \mathbb{1}$$

$$\mathbb{1} \in \text{Span}\{\psi_+, \psi_-\}$$